

InventOR: A Prompt-Configured, No-Fine-Tuning LLM Workflow for Material-Level Inventory Control

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Abstract

Material planners in manufacturing set reorder points, safety stocks and order quantities for thousands of materials, usually with ERP parameters that were last revised long ago. We present *InventOR*, a human-reviewed decision aid in which a large language model (LLM), configured only through a prompt and without any training or fine-tuning, reads one material’s demand and lead-time history and proposes a reorder-point/order-quantity (r, Q) policy. Deterministic software parses every proposal, checks it against minimum order quantities and storage limits, and replays it on held-out post-cutoff demand before a planner sees it. We evaluate the aid on 365 plant-material pairs from three plants of an industrial manufacturer (346 in the common-feasible cohort), with three independent LLM runs. Against ERP-derived parameters, the LLM policies cost about 11% less in the simulation at a service level 0.1–0.4 percentage points lower; against an operations-research (OR) policy that uses the same ERP inputs the cost is about 30% lower. However, an OR policy computed from the history alone costs about 6% *less* than the LLM at the same service, so much of the apparent gain reflects the

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weakness of the ERP benchmarks rather than an optimization advantage of the LLM; the saving comes from holding about 17% less inventory, at about 13% more stockout days. When shortages are priced at one unit cost or more, the ERP-derived parameters become the cheapest arm and the LLM is about 2.4% more expensive than them. When the ERP planning parameters are withheld from the LLM in a 365-pair ablation, it stays about 9% cheaper than ERP-derived parameters and about 9% more expensive than the history-only OR. Storage limits that appear satisfied in a static check are exceeded on 53–66% of pairs during unconstrained simulated trajectories, depending on the method; a hard-limit replay is therefore our primary evaluation, and it leaves the ranking unchanged. At one plant the LLM is 31–36% more expensive than ERP-derived parameters in the three full runs, a result that is robust to our calendar repairs. We derive rules for when such a tool should be trusted, reviewed or rejected, and set out a research agenda.

Keywords: Generative Artificial Intelligence, Large Language Models, Operations Research, Supply Chain Management, Decision Support Systems, Inventory Control

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point that triggers an order, a safety stock that absorbs demand and lead-time variability, and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand patterns, supplier performance and storage constraints change. Revisiting them materially requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. That scarce skill is exactly what a

10 conversational large language model (LLM) with a code interpreter appears
11 to offer, and a rapidly growing literature explores such tools for supply-chain
12 and operations decisions [1, 2, 3, 4, 5].

13 The promise is real, but so are three problems that limit its use in prac-
14 tice. First, LLM outputs can be unfaithful or fabricated [6], so a proposed
15 policy must be treated as a candidate and not a decision. Second, claims of
16 superiority over ERP parameters are hard to interpret when the LLM and
17 the comparators are given different information; an LLM that sees a plant’s
18 ERP reference values and is compared with an OR policy that does not, or
19 the reverse, tells us little about the LLM. Third, inventory policies live inside
20 physical limits: a storage rule that holds when the policy is written down
21 can be violated as stock builds up over time.

22 This paper addresses all three in an industrial application. We describe
23 and evaluate *InventOR*, a decision aid built around an enterprise LLM appli-
24 cation that is configured with a prompt and receives no project-specific train-
25 ing or fine-tuning. The LLM proposes one (r, Q) policy per plant-material
26 pair; deterministic software parses, checks and simulates each proposal; a
27 planner reviews proposals and simulation results before any ERP parameter
28 is changed. Our contributions are:

29 **C1. A human-reviewed decision-aid design** (Section 3) that separates
30 generation, deterministic gating, simulation and planner review, and
31 states explicit rules for when its output is trusted, escalated or rejected.

32 **C2. An information-controlled evaluation** (Section 4) with four refer-
33 ence arms that differ in the information they use and, partly, in method,
34 including an OR policy computed from history alone and an LLM abla-
35 tion that is denied the ERP planning parameters. This separates what

36 the LLM adds from what it is given.

37 **C3. Trajectory-level storage verification:** every policy is replayed with
38 orders truncated at the storage limit, and the replay, not the static
39 parameter check, is the primary result.

40 **C4. Complete-cohort evidence from three LLM runs on 365 pairs,**
41 with plant-level, regime-level and sensitivity analyses, and managerial
42 and research implications.

43 Our main findings are as follows. Against ERP-derived parameters, the
44 LLM policies are about 11% cheaper at a service level that is 0.1–0.4 points
45 lower. That advantage largely disappears when a reasonable OR baseline is
46 available: a history-only (r, Q) policy is about 6% cheaper than the LLM at
47 essentially equal service, and the LLM’s saving over ERP parameters comes
48 from carrying about 17% less inventory at about 13% more stockout days;
49 once shortages are priced at one unit cost or more, the ERP-derived pa-
50 rameters are the cheapest arm. The LLM’s value is therefore not superior
51 optimization but the ability to produce a reviewable, auditable policy for
52 every material without writing code. Storage limits are violated along un-
53 constrained simulated trajectories for 53–66% of pairs under every method,
54 which makes trajectory checks a prerequisite for any replenishment decision
55 aid. At one plant the LLM is 31–36% more expensive than ERP-derived
56 parameters, which we use to derive guidance on where to trust and where to
57 review.

58 The remainder of the paper is organized as follows. Section 2 positions
59 the work in the literature. Section 3 explains the decision aid for a non-
60 specialist reader. Section 4 describes the data and the evaluation design.

61 Section 5 reports results. Section 6 derives managerial insights, and Sec-
62 tion 7 discusses limitations and research directions. Formal definitions are
63 in Appendix Appendix A.

64 **2. Related Work and Positioning**

65 *LLMs in operations and supply-chain decisions..* Reviews and frameworks
66 map the opportunities for generative AI in supply chains [1, 2, 3, 4, 7] and
67 stress that adoption depends on assigned human roles, data quality and gov-
68 ernance [5, 8, 9, 10]. A recent conceptual line proposes agentic digital twins
69 that combine AI agents with model-based simulation and optimization to
70 support decisions [11], and an LLM-plus-digital-twin method has been built
71 for closed-loop supply-chain operations [12]. Our decision aid follows the
72 same logic at a smaller scale: an LLM proposes and a simulation model
73 checks. Most technical LLM work focuses on translating natural-language
74 problem statements into optimization models and solver code [13, 14, 15, 16],
75 on multi-agent control and consensus-seeking in inventory flows [17, 18], on
76 extracting supply-network structure from text [19, 20], on disruption mod-
77 elling [21], or on evaluating LLMs on classical operations-research tasks such
78 as vehicle routing [22]. The closest LLM study to ours uses an LLM as a
79 forecasting copilot for spare-parts demand, combined with simulation, and
80 reports high variability for irregular demand that can be reduced by averag-
81 ing repeated queries and enriching prompts [23]. We share the concern about
82 run-to-run variability and test it directly (Section 5.7), but differ in that the
83 LLM proposes a complete replenishment policy rather than a forecast.

84 *Data-driven inventory control without LLMs..* Learning-based replenish-
85 ment is well established. Deep reinforcement learning has been applied to

86 omnichannel and multi-echelon systems [24, 25] and aerospace replenishment
 87 [26]; deep-learning forecasting has been integrated with replenishment [27];
 88 contextual distributionally robust optimization uses side information to
 89 set order quantities [28]; and distribution-free online algorithms handle
 90 capacitated warehouses under uncertain price and demand [29]. Robust
 91 inventory control under demand and lead-time uncertainty has a longer
 92 history [30, 31, 32], and lead-time variability and dependence are known
 93 to propagate through ordering decisions [33], while safety stock and safety
 94 lead time are alternative buffers against unreliability [34]. The closest
 95 operational neighbour is a trained data-driven framework that jointly
 96 sets replenishment parameters, safety stock and safety lead time on real
 97 automotive data [35]. These methods require model training, historical
 98 scenario generation or tuned hyperparameters for each setting; InventOR
 99 requires none, at the price of an LLM-based generator whose internal logic
 100 cannot be inspected beyond its written justification.

101 *Intermittent and spare-parts demand.* Material histories in manufacturing
 102 are frequently intermittent or lumpy. The statistical literature offers special-
 103 ized estimators and classification schemes [36, 37, 38, 39, 40], and recent stud-
 104 ies develop feature-based forecast combinations and temporal aggregation for
 105 intermittent demand [41, 42], forecasting for spare parts across product life-
 106 cycle phases [43], and integrated forecasting and planning or multi-echelon
 107 stocking models for repairable and spare parts [44, 45]. Our regime analysis
 108 (Section 5.6) shows where an LLM that is forbidden from using specialized
 109 intermittent-demand estimators behaves differently on lumpy materials.

110 Table 1 summarizes these streams and states how the contribution of
 111 InventOR differs. InventOR is not a new inventory model and does not claim

112 to beat trained or optimized methods. It contributes (i) a way of using an off-
113 the-shelf LLM as a bounded policy generator inside a reviewed process, (ii)
114 an evaluation that controls the information available to each arm, and (iii)
115 industrial evidence on cost, service and storage feasibility. The (r, Q) policy
116 it uses is the standard workhorse of continuous-review control [46, 47, 48].

Table 1: Literature positioning of InventOR relative to recent work on AI-supported inventory and supply-chain decisions.

Study	Approach	Decision	Relation to InventOR
<i>LLMs for operations and supply chains</i>			
Jackson et al. [1]; Fosso Wamba et al. [2]	Framework / review	Broad supply-chain use	Motivates LLM use; no inventory-policy evidence
Song et al. [3]; Panja et al. [4]	Systematic reviews	LLMs in SCM and logistics	Place InventOR in the emerging empirical gap
Trifone et al. [5]; Cannas et al. [8]; Venkatesh et al. [9]	Adoption studies, search agenda	AI roles, barriers, data quality	Support human-in-the-loop design and review rules
Ivanov [11]; Le et al. [12]	Agentic LLM digital twins	Model-based decision support	Pair AI proposals with simulation; InventOR applies this to (r, Q) parameters
Nakhost et al. [23]	LLM + simulation	Spare-parts forecast	Closest LLM study; forecast, not policy
Jannelli et al. [18]; Quan and Liu [17]	Multi-agent LLM	Sequential ordering, consensus	Per-period agent decisions in simulation
AlMahri et al. [19]; Zheng and Brintrup [20]	LLM + knowledge graphs	Supply-network visibility	LLM for information extraction, not inventory control
Hamdan and Brahimi [22]	LLM benchmark	Routing	Evaluates LLM on a classical OR task
Li et al. [13]; Huang et al. [14]; Ahmadi et al. [15]; Zhang et al. [16]	NL→model solver	General OR models	Model formulation; not a fixed policy family
<i>Data-driven inventory control without LLMs</i>			
Gabellini et al. [35]	Trained data-driven	Replenishment, safety stock and safety lead time	Closest OR neighbour; needs training; automotive data
Kolyaei et al. [24]; Stranieri et al. [25]; Wang et al. [26]	Deep reinforcement learning	Replenishment	Learned policies; require training environments
Ma [27]; Wang et al. [28]; Zhang et al. [29]	Deep learning / contextual DRO / online algorithm	Forecast-plus-order; capacitated replenishment	Training, tuning or worst-case design per setting
<i>Intermittent and spare-parts demand</i>			
Sanguri et al. [41]; Li et al. [42]; Goldsmith and Sachs [43]	Statistical / ML forecasting	Intermittent demand	Specialized estimators the LLM is not allowed to use
Babaveisi et al. [44]; Caserta and D'Angelo [45]	Optimization	Spare-part stocking	Multi-echelon or repairable settings; outside our single-echelon scope
Prak et al. [39]; Babai et al. [40]	Estimation / review	Inventory under intermittency	Regime-aware benchmarks for future work
InventOR (this paper)	(this Prompted LLM + deterministic gate and simulation)	(r, Q) per plant-material pair	No training; reviewed; information-controlled evaluation

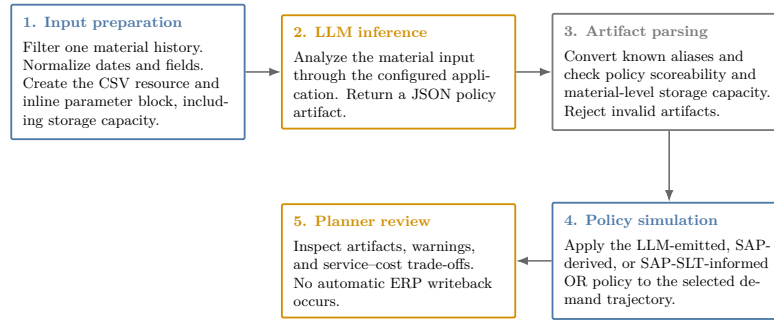
117 3. The InventOR Decision Aid

118 3.1. What the tool does, in plain terms

119 A planner’s question is: *given what this material has done in the past,*
 120 *how much safety stock should I hold, at what stock level should I reorder,*
 121 *and how much should I order?* InventOR answers this question in four steps
 122 (Figure 1).

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

123 **Step 1. Generate.** The LLM, accessed through an enterprise application
 124 with a code interpreter, receives one material’s history (daily de-
 125 mand, realized lead times, working-day flags) and a short param-
 126 eter block (unit cost, holding rate, ordering cost, MOQ, service
 127 target, storage limit, and current ERP safety stock and safety
 128 lead time for comparison). A written prompt (the only configu-
 129 ration; no weights are changed) [49] instructs it to compute de-
 130 mand and lead-time statistics, classify the demand pattern, ap-
 131 ply an economic-order-quantity (EOQ) argument and a normal-

132 approximation safety stock, respect the storage limit, and report a
133 structured policy together with a short justification and escalation
134 flag.

135 **Step 2. Gate.** Deterministic code extracts the policy, rejects malformed
136 or infeasible outputs (missing or non-numeric fields, non-positive
137 reorder point or order quantity, negative safety stock, reorder point
138 below safety stock, safety stock plus order quantity above the stor-
139 age limit) and retries the generation up to three times. The backtest
140 applies one further admission check, that the order quantity is not
141 below the MOQ.

142 **Step 3. Simulate.** Each accepted policy is replayed on the material’s de-
143 mand observed *after* the data cutoff, a held-out evaluation in the
144 sense of out-of-sample forecast testing [50, 51], together with the
145 comparator policies, producing holding and ordering cost, fill rate
146 and stockout days. This plays the role that model-based simulation
147 plays in digital-twin decision support [11], at the scale of a single
148 policy.

149 **Step 4. Review.** A planner sees the policy, the LLM’s justification and
150 flag, and the simulated comparison. Only the planner changes ERP
151 parameters.

152 The role of the LLM is deliberately narrow. It is a *candidate generator*
153 whose output is never trusted without the gate, and it is not used to fore-
154 cast, to place orders, or to run the simulation. The prompt prohibits Monte
155 Carlo simulation, distribution fitting, machine-learning libraries and special-
156 ized intermittent-demand estimators, so that the policy rests on transparent

157 statistics a planner can reproduce.

158 3.2. *Trust, review and reject rules*

159 We propose three explicit rules for planners, in line with guidelines for
160 human–AI interaction and with internal-auditing practice for AI systems
161 [52, 53], and motivated by the evidence in Section 5.

162 **Trust (apply after review)** when the proposal passes the gate, the sim-
163 ulated trajectory respects the storage limit when orders are truncated at it,
164 and the simulated cost is not above that of a history-only OR reference with
165 a fill rate not more than about 0.5 points lower.

166 **Escalate** when the LLM raises an escalation flag, the material is lumpy or
167 has strongly seasonal or trending demand, the plant or calendar data were
168 repaired, or the policy differs strongly between repeated generations.

169 **Reject** when the proposal fails the gate after the retry budget, exceeds the
170 storage limit in the hard-limit replay, or is dominated (higher cost and lower
171 fill) by the history-only reference.

172 3.3. *Provenance and reproducibility of the configuration*

173 The system prompt, which fully defines the task, the forbidden methods,
174 the capacity rules and the output schema, is a plain-text file that is deployed
175 unchanged in the enterprise application; it is distributed with the paper
176 (Supplementary File S1) and the code repository. Every call logs the SHA-
177 256 of its input data, the UTC time, the application session identifier and
178 the full raw response. The platform does not expose the underlying model
179 identifier or sampling settings to the client; the evaluated model (Anthropic
180 Claude Sonnet 4.6, with a code interpreter) is therefore declared by the

181 operator and is not machine-verifiable. We report this limitation openly and
182 base the reproducibility argument on the prompt file, the per-call logs and
183 three independent runs (Section 5.7) rather than on a model checkpoint.

184 4. Application and Evaluation Design

185 4.1. Industrial application and data

186 The application is a pilot at a large manufacturer. The dataset contains
187 233,078 daily records for 411 plant-material pairs from three plants (A, B
188 and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon
189 observed to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring
190 (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For
191 19 pairs (all in Plant A) the supplied MOQ plus the reference safety stock
192 exceeds the storage limit, so no comparator can satisfy both rules; these are
193 excluded from outcome comparison, leaving a *common-feasible cohort* of 346
194 pairs (215, 65 and 66). All 365 pairs receive an LLM policy. Demand is
195 concentrated: the five largest materials account for 40.3% of demand and
196 the ten largest for 57.8%. The source working-day flag of Plant C is all zero;
197 we reconstruct it from the same-date majority of Plants A and B (ties count
198 as working days) and test the effect of this repair in Section 5.8. Such data-
199 quality problems are a known threat to analytics-based decision support [54]
200 and, for inventory systems, to record accuracy [55, 56, 57].

201 4.2. Reference arms and information content

202 The LLM and the classical comparators do not necessarily use the same
203 information, and this matters for the interpretation of any cost difference,
204 because ERP safety stock and safety lead time embed accumulated planner
205 judgement, and safety stock and safety lead time are alternative ways of

Table 2: Fields of the material-scoped CSV given to the LLM.

Column	Type	Description and use
date	date	Observation date; time ordering and aggregation
actual_demand	numeric	Realized daily demand; mean, variability, intermittency, regime
delivery_time	numeric	Recorded replenishment lead time (receipt rows)
is_working_day	binary	Working-day flag; demand statistics over working days
minimum_order_quantity	numeric	MOQ; order-size constraint
unit_cost, holding_cost_rate, ordering_cost	numeric	Unit value, annual carrying-cost rate, fixed ordering cost; EOQ and cost evaluation
service_level_target	numeric	Target service used in safety-stock computation
max_storage_units	numeric	Storage limit: $MOQ \leq Q$ and $SS + Q \leq \text{max_storage_units}$
safety_stock_units, safety_time_workdays_actual	numeric	ERP planning parameters (current safety stock, safety lead time); withheld in the ablation arm
reliability_calc, supplier_plant_distance, transportation_network	mixed	Supplier and lane context

206 buffering unreliability [34]. Planned lead times can also drift upward when
 207 they are adjusted dynamically (the “lead time syndrome” [58]), which would
 208 inflate any comparator that adds them to the lead time. We therefore con-
 209 struct arms that make the information explicit (Table 3).

Table 3: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP used	inputs	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock		Status quo of the plant
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time		Original OR comparator
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both		OR arm matched to the LLM’s full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None		ERP-independent baseline
LLM (runs 1–3)	LLM proposal with all fields of Table 2	Both		Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to “not supplied”	None		Ablation: what the LLM does when it cannot anchor on ERP

210 4.3. Simulation, capacity replay and metrics

211 Each pair is simulated independently on the post-cutoff horizon as a
 212 single-item, single-echelon continuous-review (r, Q) system [46, 59, 60, 48, 47]
 213 with lost sales [61], in the spirit of simulation-based experiments on supply
 214 and timing uncertainty [62], a shared opening stock, an empty pipeline and
 215 arrivals after the rounded mean lead time in working days (Appendix Ap-
 216 pendix A). MOQs are treated as hard order-size constraints, as imposed
 217 by suppliers [63], and storage limits as capacity bounds on the replenish-
 218 ment decision [29]. The primary metrics are the mean material-horizon
 219 holding-plus-ordering cost, the mean material-level fill rate, the aggregate
 220 (volume-weighted) fill rate and the number of stockout days. We report
 221 paired differences with 10,000-sample bootstrap 95% intervals.

222 A static check $SS + Q \leq \text{max_storage_units}$ does not show that storage
 223 limits are respected as the inventory evolves, because the reorder point in-
 224 cludes lead-time demand and inventory on order can exceed the limit when
 225 it arrives. We therefore record, for every policy and every simulated day, the
 226 post-receipt on-hand stock and report the share of pairs and days above the
 227 limit (the *unconstrained* simulation). We then *replay* every policy with each
 228 order truncated to the storage limit minus the current inventory position
 229 (the *hard-limit replay*). The hard-limit replay is our primary evaluation.

230 4.4. Repeated runs and sensitivities

231 To examine run-to-run variation, the entire cohort is generated three
 232 times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1
 233 serves as the reference for tabulated results unless stated otherwise; runs 2
 234 and 3 provide replicates. Sensitivity analyses vary the working-day recon-
 235 struction (majority, any-plant, strict-all, exclusion of Plant C and calendar-

day arrivals), the shortage penalty (zero, one and five times unit cost), and the exclusion of diagnostic outliers (52 pairs with coefficient-of-variation error above 0.75 or safety stock outside half to twice the reference).

5. Results

5.1. Completion and policy characteristics

All four runs (three with the full input and one ablation, Section 5.4) return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most three attempts per pair. Retries were mostly caused by schema errors (invalid policy), not by capacity violations. The validity gate rejects non-positive reorder points and order quantities, negative safety stocks and reorder points below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of the 346 in the cohort below), so for a majority of pairs the SAP-derived arm holds no safety stock at all.

5.2. Headline comparison with ERP-derived parameters

Unless stated otherwise, all results from here on use the hard-limit replay (every order truncated at the storage limit) on the 346-pair common-feasible cohort, with working-day receipts and no shortage penalty; Section 5.5 reports the unconstrained simulation for comparison. Table 4 compares all arms. In run 1 the LLM policies cost 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed OR comparator (mean 2,928 versus 3,305 and 4,211). The mean fill rate is 0.18 points lower than for SAP-derived (98.11% versus 98.30%; bootstrap interval -0.54 to $+0.10$) and 0.11 points lower than for the SLT-informed OR. The mean cost difference against SAP-derived is -377 (95% interval -651 to -154); the LLM is cheaper than SAP-derived in 241 of 346 pairs and, in 226 pairs, simultaneously no more

expensive and no worse in fill. The LLM incurs more stockout days in total (910 versus 804 and 831), and aggregate fill is 78.87% versus 79.34% and 79.15%. Aggregate fill is dominated by a few high-demand materials: in the unconstrained simulation of run 1, excluding the five largest materials raises all three to 94.6–95.8% (Figure 2).

The cost decomposition shows where the saving comes from. Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and has about 16% lower holding cost, while it places about the same number of orders per year (17.3 versus 17.6). The saving is therefore a trade of inventory for service: the LLM holds less stock and accepts about 13% more stockout days. It is not the consequence of a better ordering rhythm, and it is not free of service cost. Section 5.8 shows that, once shortages carry a cost, this trade no longer favours the LLM.

Table 4: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year.

Arm	Mean cost	Holding	Ordering	Avg. inventory	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days
SAP-derived	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890

5.3. Information asymmetry: what the LLM adds beyond the ERP inputs

Table 5 compares the arms in the hard-limit replay on the 346 pairs of the cohort, for all three runs. Three results stand out.

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248). The ERP safety lead time inflates their effective lead time, hence their buffers; against them the LLM is about 30–31% cheaper in every run. This comparison therefore mainly measures how heavily ERP inputs over-buffer, not the quality of the LLM.

Demand concentration and fill sensitivity

Largest-demand materials strongly influence full-cohort aggregate fill.

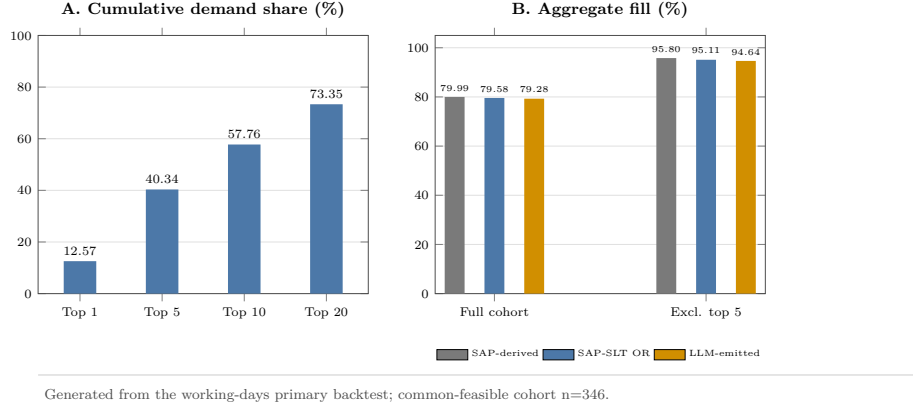


Figure 2: Demand concentration and aggregate-fill sensitivity (unconstrained simulation, run 1). Panel A: cumulative demand share of the largest materials. Panel B: aggregate fill for the full cohort and after excluding the five largest materials.

Second, the *history-only OR*, which uses no ERP inputs, has the lowest cost of all non-LLM arms (2,764 at a fill of 98.14%). Against it the LLM is 5.9–6.0% more expensive in all three runs (run 1: mean difference +164, 95% interval +91 to +240), with fill differences of -0.03 , -0.25 and $+0.01$ points whose intervals include zero in all three runs. The LLM is cheaper than the history-only OR in only 74–77 of 346 pairs. Part of the LLM’s lower cost relative to ERP therefore reflects the weakness of the ERP benchmarks and not LLM superiority.

Third, the LLM is nonetheless cheaper than ERP-derived parameters (11.3–11.4%) at a mean fill that is lower by 0.14–0.40 points (the interval excludes zero only in run 2), so it is useful when the alternative is the status quo, and it approaches a competent OR benchmark without code or training. Whether the LLM relies on the ERP planning parameters it is given is tested in the ablation of Section 5.4.

Table 5: Hard-limit replay: mean cost and mean fill by arm, $n = 346$; LLM cost relative to each arm (negative = LLM cheaper).

Arm	Mean cost	Mean fill (%)	LLM cost vs arm (%)		
			Run 1	Run 2	Run 3
SAP-derived	3,305	98.30	-11.4	-11.3	-11.4
SLT-informed OR	4,211	98.23	-30.5	-30.4	-30.5
ERP-floor OR	4,248	98.23	-31.1	-31.0	-31.1
History-only OR	2,764	98.14	+5.9	+6.0	+5.9
LLM (runs 1, 2, 3)	2,928 / 2,931 / 2,929	98.11 / 97.89 / 98.16			
LLM without ERP (ablation)	3,014	98.06	-8.8 [†]		

[†] Relative to SAP-derived; other comparisons are in Section 5.4.

5.4. LLM without ERP planning parameters (ablation)

In this ablation the LLM receives exactly the same material histories, cost parameters, MOQ, service target and storage limit as in runs 1–3, but the ERP safety stock and ERP safety lead time are blanked in the uploaded CSV and marked “not supplied” in the inline parameter block; the system prompt file is unchanged. It tests whether the LLM anchors on the ERP planning parameters it is given, and therefore whether the comparison with the ERP-informed arms is confounded by information. The ablation run has the same structure as runs 1–3: all 365 pairs returned a valid, storage-compliant policy.

Policy values. Withholding the ERP parameters changes the LLM’s policies only slightly beyond its ordinary run-to-run variation. The median ratios of safety stock and reorder point to run 1 are both 1.00 (346 common pairs). The reorder point differs by more than 10% in 54.6% of pairs, against 48.3% between runs 1 and 3 and 48.0% between runs 2 and 3 with the ERP parameters supplied; the safety stock differs by more than 10% in 11.0% of pairs, against 9.8% and 9.5%. The median absolute reorder-point difference is 128 units, against 90 and 84 between ERP-supplied runs. The mean ratio is slightly above one (1.04 for safety stock, 1.05 for the reorder point), so the LLM tends to buffer a little more when it has no ERP anchor.

316 *Outcomes.* In the hard-limit replay the LLM without ERP parameters
 317 has a mean cost of 3,014 at a mean fill of 98.06%, against 2,928–2,931 and
 318 97.89–98.16% in runs 1–3. It remains 8.8% cheaper than SAP-derived pa-
 319 rameters (95% interval of the mean difference -566 to -61 ; fill -0.24 points,
 320 interval including zero) and 28.4–29.1% cheaper than the SLT-informed and
 321 ERP-floor ORs, but it is 9.0% more expensive than the history-only OR
 322 (interval $+134$ to $+387$; fill -0.09 points; cheaper in 91 of 346 pairs). In the
 323 unconstrained simulation the corresponding figures are -7.0% against SAP-
 324 derived (interval -513 to $+16$, which includes zero), -23.2% and -23.8%
 325 against the two ERP-informed ORs, and $+8.8\%$ against the history-only OR.
 326 Withholding the ERP parameters thus raises the LLM’s cost by roughly 3
 327 percentage points relative to the history-only OR and does not reverse any
 328 ranking.

329 *By plant.* The pattern is the same as with ERP inputs: the LLM remains
 330 cheaper than SAP-derived at Plants A (215 pairs, -10.8%) and B (65 pairs,
 331 -21.9%) and more expensive at Plant C (66 pairs, $+41.6\%$ against $+36.2\%$
 332 in run 1; cheaper in 8 pairs). The Plant C loss is therefore not caused by
 333 the LLM leaning on ERP parameters either.

334 *Interpretation.* Two conclusions follow. First, the information-
 335 asymmetry concern (that the LLM benefits from ERP inputs) does not
 336 explain the LLM’s performance: with the ERP parameters removed the
 337 LLM keeps most of its advantage over ERP-derived parameters, and the
 338 gap to the history-only OR widens only moderately. Second, the ERP
 339 parameters are of limited use to the LLM; most of its output is determined
 340 by the demand and lead-time histories. A practical consequence is that the
 341 LLM can be deployed for materials whose ERP planning data are missing or
 342 stale. One limitation is that the ablation tests only the two ERP planning

343 fields; the LLM still sees unit cost, holding rate, ordering cost, MOQ, service
344 target and storage limit, which all reference arms also use.

345 5.5. *Storage limits along simulated trajectories*

346 All 365 LLM policies satisfy the static rule $SS+Q \leq \text{max_storage_units}$.
347 Along the simulated trajectories, however, the unconstrained simulation
348 exceeds the limit on post-receipt on-hand stock for 183 pairs (52.9%) for
349 the LLM (10.2% of simulated pair-days; median peak 1.02 times the limit;
350 maximum 2.52), 198 pairs (57.2%; 15.9% of days) for SAP-derived and 230
351 pairs (66.5%; 23.4% of days) for the SLT-informed OR; no pair starts above
352 the limit. The three LLM runs give 183, 181 and 178 pairs. Exceedances
353 arise because inventory on order and lead-time demand can add to stock
354 above $SS+Q$; the static check is therefore necessary but not sufficient, and
355 it applies to every method, not just the LLM.

356 Under the hard-limit replay no arm exceeds the limit by more than one
357 unit (rounding). Enforcing the limit raises the LLM mean cost from 2,850
358 to 2,928 and lowers its mean fill by 0.04 points (98.15% to 98.11%); SAP-
359 derived rises from 3,184 to 3,305 and the SLT-informed OR from 3,852 to
360 4,211, with the latter placing 24.5 instead of 15.8 orders on average. The
361 ranking among arms is unchanged, and the LLM’s relative advantage over
362 SAP-derived is 11.4% instead of 10.5%. We therefore use the replay as the
363 primary evaluation.

364 5.6. *Plants and demand regimes*

365 The average hides a strong plant effect (Table 6, hard-limit replay, run 1).
366 The LLM is 14.1% cheaper than SAP-derived in Plant A (cheaper in 179 of
367 215 pairs) and 22.7% cheaper in Plant B (54 of 65), but 36.2% *more* expensive

in Plant C, where it is cheaper in only 8 of 66 pairs. The Plant C loss is similar in the other full runs (+32.2% and +31.5% in runs 2 and 3) and larger without the ERP parameters (+41.6%). Plant C is also the plant whose calendar we repaired; Section 5.8 shows that the loss persists under every working-day reconstruction in which Plant C can be scored. Against the history-only OR the LLM is 7.2% costlier in Plant A, 3.6% in Plant B and 6.7% in Plant C (run 1), so no plant favours the LLM over a plain history-based (r, Q) policy. Note that the SLT-informed and history-only ORs give identical results in Plant C, because the ERP safety lead time is zero for all 66 Plant C pairs.

By demand regime, the 33 lumpy materials show the largest cost reduction against SAP-derived (−25.4% in run 1; −23.7% to −25.2% in the other runs), whereas the 313 smooth materials show −9.8% at a 0.24-point lower mean fill (98.64% versus 98.88%). The service effect for lumpy materials is not consistent (mean fill +0.38 points in run 1, −0.29 in run 2), and the history-only OR is again 9% cheaper than the LLM there while achieving a much lower aggregate fill (54.3% versus 57.0% for the LLM and 65.8% for SAP-derived), so part of any cost saving on lumpy materials is bought with service. With only 33 lumpy materials we regard this as indicative only. By demand variability (terciles of the coefficient of variation), the LLM is 15.7% cheaper than SAP-derived at equal fill in the lowest tercile, 7.5% cheaper with a 0.68-point lower mean fill in the middle tercile (98.10% versus 98.78%), and 12.9% cheaper in the highest tercile.

Table 6: Mean simulated cost per pair and mean fill by plant, run 1, hard-limit replay (common-feasible cohort). The SLT-informed and history-only ORs coincide in Plant C.

Plant (<i>n</i>)	LLM cost	SAP-derived cost	SLT-informed OR cost	History-only OR cost	LLM vs SAP	Cheaper pairs	Fill LLM / SAP (%)
All (346)	2,928	3,305	4,211	2,764	−11.4%	241	98.11 / 98.30
A (215)	2,391	2,782	3,779	2,231	−14.1%	179	99.18 / 99.37
B (65)	4,916	6,360	7,329	4,746	−22.7%	54	96.23 / 96.47
C (66)	2,719	1,997	2,549	2,549	+36.2%	8	96.50 / 96.61

391 5.7. Run-to-run stability

392 Across the three runs the aggregate results are very similar: LLM cost
 393 relative to SAP-derived is -10.5% , -9.8% and -10.2% in the unconstrained
 394 simulation, and -11.4% , -11.3% and -11.4% in the hard-limit replay, so
 395 the spread is below one percentage point. The distributions of policy values
 396 are near-identical (median reorder point 1,584, 1,582 and 1,466; median or-
 397 der quantity 2,737, 2,777 and 2,693; median safety stock 683, 688 and 669;
 398 Figure 3). Individual policies, however, vary. Comparing runs 2 and 3 on
 399 the 365 pairs, all three policy values are identical for only 36 pairs; every
 400 policy resolves to the (r, Q) family, and the median absolute differences are
 401 81 units for the reorder point, 6 for the order quantity and 3 for the safety
 402 stock. A few pairs vary strongly (for one material the reorder point was
 403 14,882, 14,947 and 29,218 units in the three runs; for two others the safety
 404 stock was zero in one run and 660 or 1,075 units in the others). The practical
 405 implication is that one generation per material is a noisy estimate whose ag-
 406 gregate behaviour is stable but whose individual values should be reviewed.
 407 Non-determinism of this kind is a documented property of LLMs [64], so
 408 that repeated-run reporting should be standard.

409 5.8. Sensitivity analyses

410 *Plant C calendar..* Plant C’s source working-day flag is all zero, which re-
 411 quired reconstruction. We re-scored all policies (hard-limit replay) under
 412 four alternatives: any-plant, strict-all, exclusion of Plant C and calendar-
 413 day arrivals. In every variant in which Plant C can be scored (any-plant and
 414 calendar-day arrivals), the LLM remains 31–35% more expensive than SAP-
 415 derived in Plant C across the three full runs. The cohort-wide advantage
 416 over SAP-derived is -10.8% to -11.1% in these variants (primary calen-

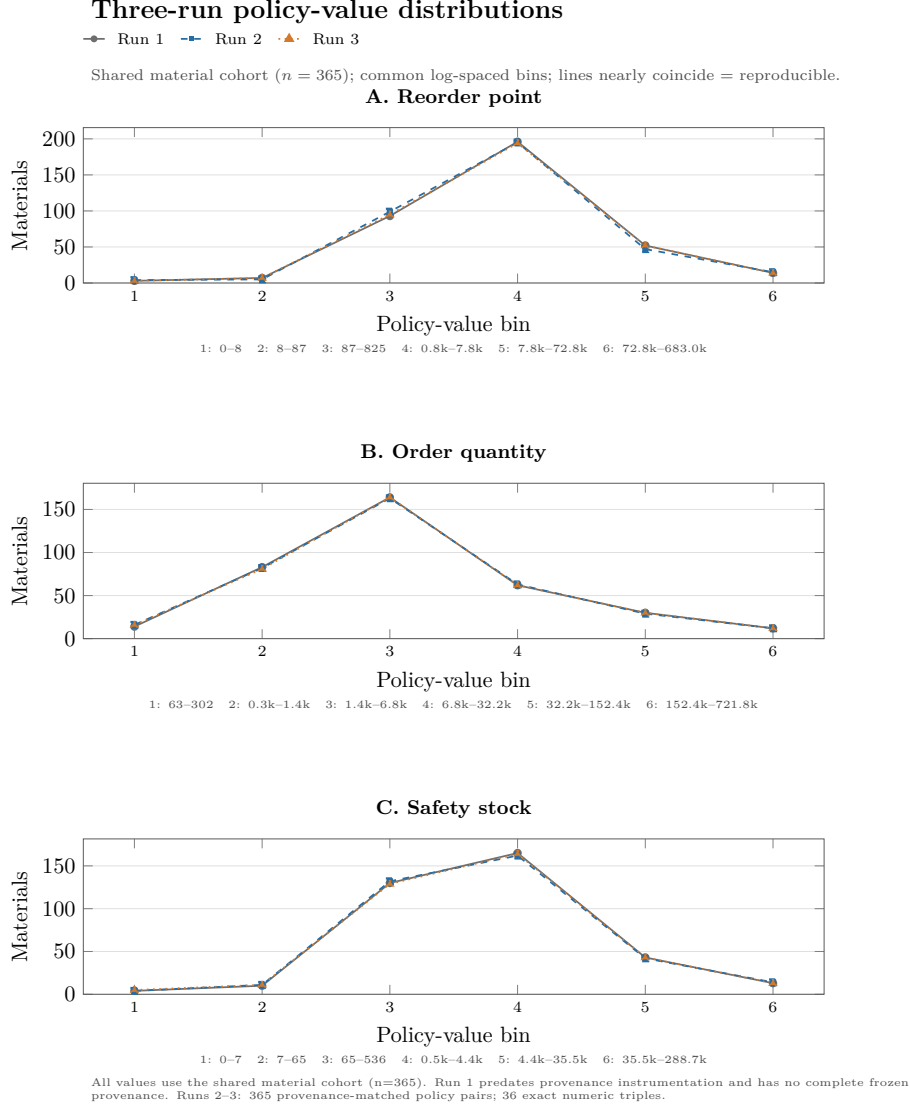


Figure 3: Policy-value distributions across the 365 shared pairs for runs 1, 2 and 3 (materials per common log-spaced bin); near-coincident lines indicate that aggregate behaviour is reproducible.

417 dar: -11.3% to -11.4%), and the cost excess over the history-only OR is
418 $5.6\text{--}6.1\%$. The strict and exclusion variants (280 pairs) drop Plant C and
419 show a larger LLM advantage over SAP-derived (-17.0% to -17.6%) and

a 5.8–6.6% excess over the history-only OR. The Plant C result is therefore not an artefact of our calendar repair. One residual risk remains: the LLM saw demand series for Plant C that were themselves built on the repaired calendar, and only a re-generation with alternative inputs could remove that dependence.

Shortage penalty.. With a positive lost-sales penalty the zero-penalty cost ranking changes. Table 7 gives mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay for penalties of zero, one and five times the unit cost. The SAP-derived parameters become the cheapest arm as soon as shortages are priced, because they carry more inventory and fewer stockouts. Run 1 of the LLM then costs 2.4% more than SAP-derived at both one and five times the unit cost; it remains cheaper than the SLT-informed OR (1.2% lower at both multiples) and the ERP-floor OR, and slightly cheaper than the history-only OR (0.8% lower at both multiples). The LLM’s cost advantage over ERP-derived parameters is therefore specific to the case in which service shortfalls are not charged, and the service deficit is a real cost when shortages are priced.

Table 7: Mean total cost per pair in the hard-limit replay (346 pairs) when lost sales are charged at a multiple of the unit cost.

Penalty	SAP-derived	SLT-informed OR	ERP-floor OR	History-only OR	LLM run 1	LLM run 2	LLM run 3	LLM no ERP
0×	3,305	4,211	4,248	2,764	2,928	2,931	2,929	3,014
1×	1,634,136	1,694,183	1,694,220	1,687,191	1,673,003	1,683,280	1,671,759	1,689,961
5×	8,157,462	8,454,072	8,454,109	8,424,899	8,353,306	8,404,676	8,347,082	8,437,748

Diagnostic outliers and calendar arrivals.. In the unconstrained simulation of run 1, removing the 52 diagnostic outliers (314 pairs; coefficient of variation error above 0.75 or safety stock outside half to twice the reference) leaves LLM, SAP-derived and SLT-informed OR aggregate fill at 79.96%, 80.38% and 79.99%, with mean cost 2,864, 3,098 and 3,748, so the ordering

442 of the arms is unchanged. The findings thus do not depend on a few extreme
443 cases.

444 6. Managerial Insights

445 1. **Position the tool as a reviewed decision aid, not an autonomous**
446 **policy engine.** Single generations differ at the material level, and
447 roughly 6–8% of outputs carry a free-text escalation flag. The gate-
448 simulate-review structure (Section 3.2) captures most of the value of the
449 LLM (a candidate policy for every material, with a justification) while
450 leaving the decision to the planner.

451 2. **Benchmark against a history-only OR policy before claiming**
452 **value.** ERP-derived parameters and ERP-informed OR policies are weak
453 benchmarks: safety stocks are often stale or zero and safety lead times
454 inflate the effective lead time. A planning team that evaluates an LLM
455 aid only against ERP will overestimate its benefit. The history-only (r, Q)
456 is cheap to compute and was cheaper than the LLM in our data.

457 3. **Where the LLM is useful.** Plants A and B (cost reductions of 14–23%
458 against ERP), lumpy materials (about 25% cheaper than ERP-derived
459 parameters, although with mixed service effects and only 33 materials)
460 and the many materials with no analyst time are the sweet spot. Plant C,
461 where the LLM loses, indicates that conditions with repaired calendars,
462 small samples or atypical lead-time behaviour need escalation.

463 4. **Verify storage along the trajectory.** The static rule $SS + Q \leq \text{limit}$
464 cannot protect a warehouse: 53–66% of pairs breach the limit in simula-
465 tion under every method. Operations should enforce a cap in the order-

466 release logic (order-up-to or truncation), or treat the limit as a design
467 requirement in the policy.

468 **5. Price shortages explicitly.** Without a shortage penalty the cost rank-
469 ing rewards under-stocking. With a penalty of one unit cost the LLM’s
470 cost advantage over SAP disappears (it is about 2.4% more expensive).
471 Planners should set service targets and penalties before reading the cost
472 comparison.

473 **6. Freeze and log the configuration.** Treat the system prompt as a
474 controlled file, log every call (input hash, time, session, raw response)
475 and re-run before changing the prompt. Our reproducibility argument
476 rests on this log, not on the platform’s model identifier, which the client
477 cannot see.

478 **7. Cost and latency.** Each LLM generation took about 7–20 minutes,
479 whereas the OR baseline takes milliseconds. The LLM is attractive where
480 analyst time and not compute is scarce, but a deployment needs batching
481 and monitoring. Periodic regeneration as demand and lead times change
482 would also support adaptable planning under uncertainty [65].

483 **7. Discussion, Limitations and Research Perspectives**

484 *What the evidence shows..* In a three-plant industrial dataset, a prompt-
485 configured LLM generates feasible, reviewable (r, Q) policies for every mate-
486 rial. They are cheaper than ERP-derived parameters in the simulation, but
487 they do not beat a history-only OR policy. We therefore describe InventOR
488 as a way to make a competent analyst’s workflow available at scale, not as a
489 method that outperforms optimization. The information-controlled evalua-

tion is what allows this conclusion; without the ERP-independent arm, the first-cut result (-11.4% against SAP and -30.5% against the SLT-informed OR in the hard-limit replay) would have been presented as superiority.

Limitations.. (i) The data come from one manufacturer and three plants; generalization to other settings is not shown. (ii) The platform does not expose the model identifier or inference settings, so the model is operator-declared. (iii) The simulation excludes observed receipts, corrections and scrap and starts with an empty pipeline; a prospective field trial with planners is needed to assess realized benefit, which we do not claim. (iv) The cost model has zero shortage penalty in the primary results and standard holding and ordering costs; with a positive penalty the cost ranking changes. (v) The (r, Q) policy is a single-item, single-echelon family; multi-item coupling and shared capacity are outside the scope. (vi) Run-to-run variation of individual policies is substantial. (vii) Plant C required calendar reconstruction. (viii) No planner outcomes were measured: we make no claim about planner workload or decision quality. (ix) The reference arms are normal-approximation (r, Q) policies; we did not include an intermittent-demand-specific baseline (for example Croston- or SBA-based forecasts [36, 37]) or a trained data-driven or reinforcement-learning policy, so the comparison with the state of the art is incomplete, particularly for lumpy materials.

Research perspectives..

1. *Ensembling and consistency.* Aggregating several generations (median or consensus policies) may reduce the individual-policy noise while retaining the LLM’s justification; the copilot findings for spare parts [23] suggest this is promising.

- 515 2. *Hybrid generators.* Combining the history-only OR with the LLM (the
516 LLM reviewing and annotating the OR policy, or choosing among candi-
517 dates by regime), as proposed for LLM–OR integration [66], could capture
518 the OR’s cost level and the LLM’s flags.
- 519 3. *Comparison with trained methods.* A benchmark that compares prompt-
520 based generators with trained data-driven or reinforcement-learning poli-
521 cies [35, 24, 25] on identical data and information would clarify the trade-
522 off between setup effort and policy quality.
- 523 4. *Intermittent-demand and richer policy families.* Allowing order-up-to or
524 (s, S) policies and regime-specific estimators may help the lumpy and
525 Plant C pairs [39, 40].
- 526 5. *Planner-in-the-loop field trials.* Prospective studies should measure plan-
527 ner time, override rates and realized service and inventory, which our
528 simulation cannot, and should examine adoption drivers and barriers with
529 mixed-methods designs [9, 8, 5].
- 530 6. *Standards for information-controlled evaluation.* Studies of LLM decision
531 aids should report an ERP-independent baseline, an ablation that with-
532 holds the incumbent’s inputs, trajectory-level feasibility, and run-to-run
533 variation.

534 8. Conclusion

535 InventOR shows that a prompt-configured LLM can generate feasible,
536 justified inventory policies for every material of an industrial portfolio with-
537 out training, and that a deterministic gate and simulation make these propos-
538 als reviewable. On a three-plant dataset, the proposals cost about 11% less

539 than ERP-derived parameters at slightly lower service, but about 6% more
 540 than a history-only OR policy; when lost sales are priced at one unit cost
 541 or more, ERP-derived parameters become the cheapest arm. Storage limits
 542 that hold in a static check are exceeded along simulated trajectories for every
 543 method, and the LLM loses to ERP at one plant. We therefore recommend
 544 the LLM as a human-reviewed aid benchmarked against history-only OR,
 545 with trajectory-level storage checks, explicit shortage pricing and a frozen,
 546 logged configuration. These results are simulated-comparison evidence, not
 547 field evidence of improved planning outcomes.

548 **Appendix A. Formal Definitions**

549 *Reference calculation..* These are the standard textbook calculations [48, 59].
 550 For pair i , let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff
 551 working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the
 552 lead time in working days, K_i the ordering cost, h_i the annual holding cost
 553 per unit (unit cost times holding rate), z_i the standard normal quantile of
 554 the service target, and 250 the working days per year. The order quantity
 555 is $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i \cdot 250\bar{d}_i/h_i} \rceil\}$. The lead-time-demand standard
 556 deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i} \rceil$
 557 and the reorder point $R_i = \lceil \bar{d}_iL_i^{\text{eff}} + SS_i \rceil$, where L_i^{eff} is the mean lead time
 558 for the history-only OR and the mean lead time plus the ERP safety lead
 559 time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm
 560 uses the ERP safety stock and the same Q_i . All policies are projected to
 561 $SS_i + Q_i \leq \text{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

562 *Simulation..* On each post-cutoff working day, demand is served from on-
 563 hand stock, unmet demand is lost, and an order of size Q_i is placed when

564 the inventory position (on-hand plus on order) is at or below R_i and ar-
 565 rives after the rounded mean lead time in subsequent working observations.
 566 Observed receipts, corrections and scrap are ignored. The total cost is
 567 $TC_i = h_i \bar{I}_i T / 250 + K_i N_i$, where $\bar{I}_i$ is the mean on-hand stock over the
 568 T simulated working days and N_i the number of orders. Fill rate is served
 569 divided by demanded units. In the hard-limit replay an order is truncated
 570 to `max_storage_unitsi` minus the inventory position, and no order is placed
 571 if the truncated quantity is zero.

572 **Appendix B. Reproducibility Record**

573 Prompt file, runner code, evaluation scripts, aggregate outputs and per-
 574 call input hashes are distributed with the code release (Data and Code Avail-
 575 ability). The three runs and the ablation use the same prompt file; the ab-
 576 lation differs only in that the two ERP planning fields are blanked in the
 577 uploaded data and marked “not supplied” in the inline block. The deployed
 578 system prompt is the file `system_prompt.txt` in the release. The platform
 579 does not expose the model identifier or inference settings to the client; the
 580 model is therefore operator-declared (Anthropic Claude Sonnet 4.6 on an
 581 enterprise platform with a code-interpreter tool) and not independently ver-
 582 ified. For each call the release records the input data, a hash of the input,
 583 the UTC time, the session identifier and the raw response.

584 **Author Contributions**

585 Conceptualization, Methodology, Software, Formal analysis, Validation,
 586 Visualization, Writing – original draft, Writing – review and editing: Aris
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591 **Declaration of Competing Interest**

592 The author declares no known competing financial interests or personal
593 relationships that could have appeared to influence the work reported in this
594 paper.

595 **Data and Code Availability**

596 The raw operational data and material-level artifacts cannot be publicly
597 released because they contain commercially sensitive material, supplier
598 and planning information. A confidentiality-screened release package
599 containing analysis code, the anonymized variable schema, aggregate
600 evaluation results, manuscript sources and prompt files is available at
601 github.com/DressPD/inventor_public. A versioned Zenodo archive of
602 the release is available at <https://doi.org/10.5281/zenodo.22262052>. A
603 preprint of an earlier version is archived at <https://doi.org/10.5281/zenodo.22262180>.
604 Supplementary File S1 contains the system prompt used for all runs.

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611 **Declaration of Generative AI and AI-Assisted Technologies in the** 612 **Writing Process**

613 During manuscript preparation and revision, the author used OpenAI
614 GPT-5.6 and Anthropic Claude models through the OpenCode coding assis-
615 tant. They supported language editing, restructuring and redrafting of the
616 text, consistency checks, citation verification against Crossref metadata, La-
617 TeX proofing, and the writing and execution of the analysis scripts that pro-
618 duce the evidence tables (for example the capacity-trajectory, information-
619 asymmetry and sensitivity analyses). These writing and coding assistants
620 are distinct from the Anthropic Claude Sonnet 4.6 deployment that is the
621 object of evaluation in the research workflow. The author specified the anal-
622 yses, reviewed and edited all content and results after use, and takes full
623 responsibility for the content of the published article. Figures were rendered
624 deterministically from author-reviewed LaTeX and Python source; genera-
625 tive AI did not directly create or manipulate image content.

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